

Experiment 4: Selection Sort – Contest Prize Distribution

Aim

To implement the **Selection Sort** algorithm in Python to rank participants based on their scores in **descending order** and display the prize winners.

Problem Statement

In a coding contest, the participant with the **highest score** should receive the first prize, followed by the second highest, and so on. Implement this ranking using **Selection Sort** without using any built-in sorting functions.

Algorithm

Step 1: Start.

Step 2: Read the number of participants.

Step 3: Read the participant's **name** and **score**.

Step 4: For each position:

- Find the participant with the highest score in the remaining list.
- Swap with the current position if required.

Step 5: Display the participants in descending order of scores.

Step 6: Stop.

Source Code (Python)

```
main.py    Share
```

```
1 # Contest Prize Distribution using Selection Sort
2
3 n = int(input("Enter number of participants: "))
4
5 participants = []
6
7 for i in range(n):
8     name = input("Enter participant name: ")
9     score = int(input("Enter score: "))
10    participants.append([name, score])
11
12 # Selection Sort (Descending Order)
13 for i in range(n - 1):
14     max_index = i
15
16     for j in range(i + 1, n):
17         if participants[j][1] > participants[max_index][1]:
18             max_index = j
19
20     participants[i], participants[max_index] = participants[max_index], participants[i]
21
22 print("\nPrize Ranking")
23 for i in range(n):
24     print(i + 1, ".", participants[i][0], "- ", participants[i][1])
```

Output

```
Output
```

```
Enter number of participants: 4
Enter participant name: meera
Enter score: 35
Enter participant name: sachin
Enter score: 47
Enter participant name: david
Enter score: 55
Enter participant name: sana
Enter score: 87

Prize Ranking
1 . sana - 87
2 . david - 55
3 . sachin - 47
4 . meera - 35

=== Code Execution Successful ===
```

Time Complexity

Aspect	Complexity
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Time Complexity	$O(n^2)$
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Space Complexity	$O(1)$
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Result

The Selection Sort algorithm was successfully implemented to rank contest participants in descending order of scores. The program correctly displayed the prize ranking without using any built-in sorting function.